

# Bitcoin Futures and Option Markets: Searching for Completeness

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## Abstract

Cryptocurrencies have emerged in the last decade as a new asset class unlikely to disappear despite its extraordinary volatility. Futures contracts on Bitcoins were introduced in December 2017 by the Chicago Mercantile Exchange and options are being traded on crypto exchanges. Our goal in this paper is threefold: i) present the main features of cryptocurrency spot and derivative markets; ii) argue that storability of Bitcoins implies the existence of a convenience yield and infer from traded Future prices the risk neutral drift of the Bitcoin price in the continuous time Black-Scholes setting quite appropriate to continuously traded bitcoins. We use the prices of options traded on the Deribit Exchange to build the volatility smiles and skews observed at different dates of 2019 for short and long dated maturities and compare them to forward curves. Lastly, the robustness of the Black-Scholes formula allows capturing in our approach the stochastic volatility displayed by price trajectories and provides some answers to the incompleteness of the bitcoin market.

*Key words:* Futures on Bitcoin, Convenience Yield for Bitcoin, Bitcoin smiles and skews, bitcoin market incompleteness

## I. INTRODUCTION

Since the famous paper by Satoshi Nakamoto (2008) which brought to light the blockchain technology and the trading of Bitcoins, many other cryptocurrencies have been created- in particular the popular Ethereum. Without listing all of them, we need to mention that the crypto Libra envisioned by Facebook has brought new interest in all cryptos (and their derivatives) in 2019.

Financially settled Futures contracts were introduced in December 2017 by the CME and the CBOE. The latter has discontinued its contracts but ICE (Intercontinental Exchange Incorporated), the parent of the New York Stock Exchange will start trading bitcoin Futures in its Bakkt unit on September 23, 2019 after getting the green light of the US Commodities Futures Trading Commission (CFTC). For the first time, these Futures will be physically settled in Bitcoins, adding to the similarities between bitcoins and commodities. At the same time, the CME asked the CFTC to allow its clients to double their open bitcoin Futures positions, up to a monthly limit of 2,000 contracts (instead of the previous 1,000) per spot month. Cryptocurrency Exchange Binance launched on September 13 Futures written on its in-house token, Binance coin.

Call options give buyers the possibility of getting a financial exposure to Bitcoin prices while puts provide the classical protection to those owning the cryptocurrency. The exchange called Deribit gives a trading platform for options on Bitcoins with a fairly large liquidity and will be used for our option prices. Our goal in this paper is threefold: i) present some features of cryptocurrency spot markets and explain why the storage devices, old and new, ensure the existence of a convenience yield, a key parameter in the risk-neutral drift of the spot price dynamics; ii) propose for this *continuously traded* asset class a Black Scholes type approach to the valuation of options on Bitcoins; iii) use the prices quoted on the Deribit exchange to infer implied volatilities and build smiles and skews.

The remainder of the paper goes as follows: Section II presents some features of cryptocurrencies in recent years and discusses the features of Bitcoin Futures contracts already introduced and forthcoming. Section III makes the case for a continuous diffusion driving the price; its risk-neutral drift is derived by arbitrage from traded Future prices and discusses the prices of options on Bitcoins in the context of volatility uncertainty. In Section IV, smiles and skews of implied volatilities are displayed for options of different dates, together with the corresponding forward curves. Section V concludes.

## II. BACKGROUND ON BITCOINs AND REVIEW OF THE LITERATURE

### A. Some properties of Bitcoin spot prices

Bitcoins appeared in October 2008 with the famous paper under the “nom de plume” Satoshi Nakamoto (a single author or a group) advocating for “a fully peer-to-peer electronic cash system”. Bitcoin trading requires the necessary prerequisite; a blockchain, which is a secured platform and a ledger, and which is being developed for a vast array of applications unrelated to the trading of cryptocurrencies.

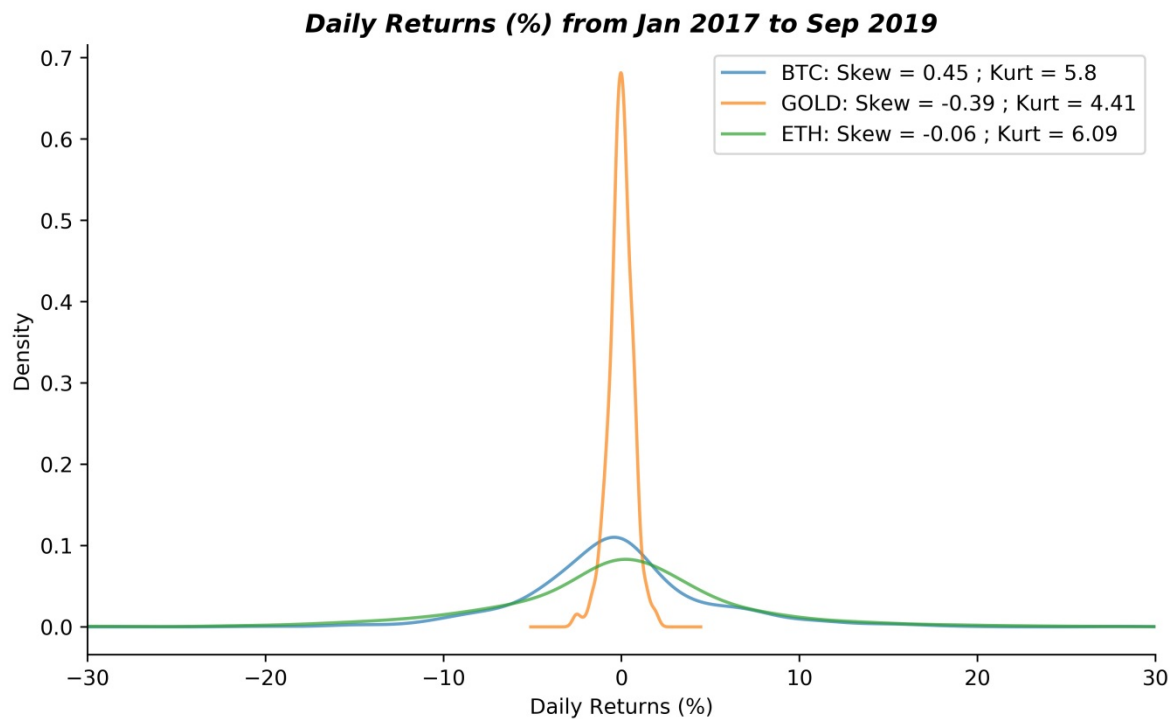
The total market capitalization of all cryptocurrencies was 828 billion dollars in January 2018 (just after two Exchanges launched futures), 200 billion in August 2018 and 272 billion in September 2019. Note that the fraction of bitcoin in the total market capitalization was 86% in January 2017, went to 34% in January 2018 and back to 68% in September 2019.

Looking at the price trajectory of Bitcoins, one can argue that the remarkable rise during the year 2017 was related to enthusiastic investors betting on an irresistible increase in its value. As of December 2017, those who had a different view were finally able to express their sentiment by taking positions in Future contracts. A cautious comparison can be made with gold which experienced a 50% drop over two years after Futures were introduced on the COMEX in December 1974; the highest value reached by the ounce of gold up to that moment was exactly the day before gold Futures started trading (see Hale et al, 2018).

Lee et al. (2018) discuss the virtues of bitcoins as an investment asset. Alexander et al. (2019) analyse spot prices of cryptocurrencies from an econometric perspective. To our best knowledge, this is the first paper which provides a tentative arbitrage approach to the valuation of bitcoin derivatives and the consistency between Futures and option prices traded on old and new Exchanges.



**Exhibit 1:** Price Trajectory of Bitcoin – April 2017 to September 2019



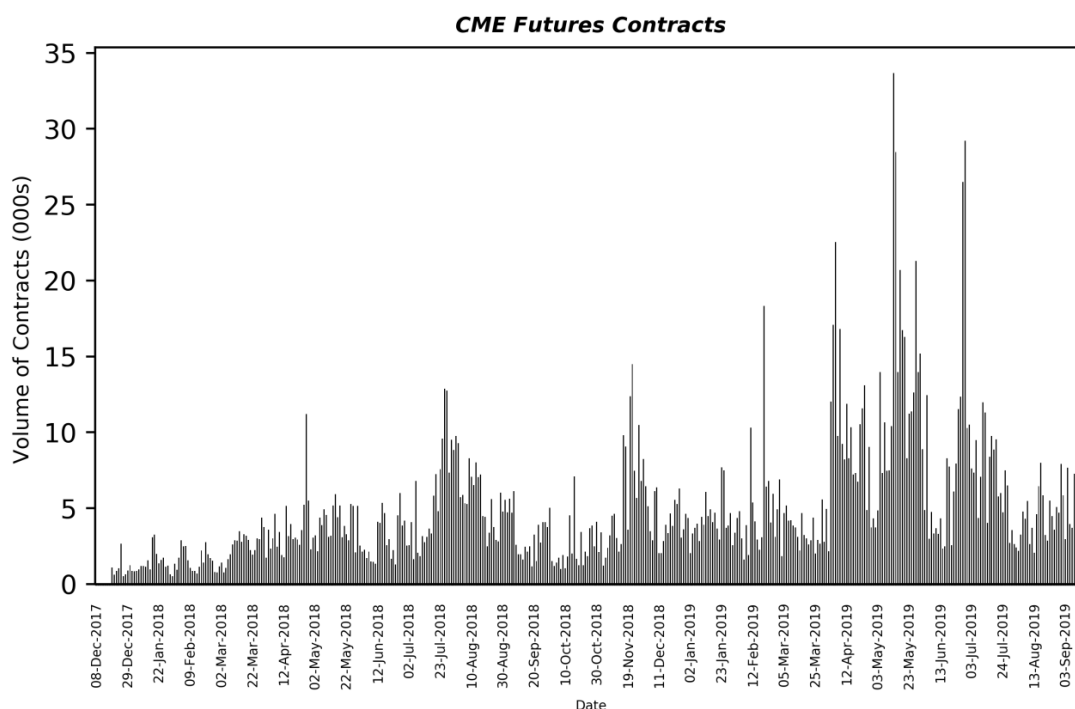
**Exhibit 2:** Daily return distributions for Bitcoin, Ethereum and Gold

### C. Futures on Bitcoin

The first group of Futures contracts was created by the Chicago Mercantile Exchange (CME) and the Chicago Board of Exchange and launched in December 2017. They have a financial settlement in dollars at maturity, like the S&P 500 Futures contracts and in contrast to the “physical” delivery of the underlying asset which prevails for most commodities. CME Futures (like CBOE Futures now delisted) are subject to significant regulatory oversight by the Commodity Futures Trading Commission (CFTC), and are protected from counterparty risk by the Clearing House of the Exchange - a desirable feature since more than 36 cryptocurrency trading venues had closed by the end of December 2018. Moreover, they partly allow avoiding the fragmentation that sometimes prevailed in the Bitcoin spot market: during periods of high volatility, prices observed for Bitcoins on different platforms over identical time periods could diverge by more than 10%.

The CME settles its contracts against an index called BRR (Bitcoin reference rate) which is an average of spot prices quoted on different exchanges. The CME contracts are related to five Bitcoins.

As of their launch in December 2017, the CME Bitcoin Futures have required an initial margin of 44% of the previous day settlement price and 40% of this number for maintenance margins. Moreover, these margins change daily (floating margins), a feature which is unique to Bitcoins like the size of the margins itself.



**Exhibit 3:** Volume of nearby Traded Futures contracts on the CME from Dec 2017 to Sept 2019

### D. Cold Storage of CryptoAssets and Coin Wallets

Cryptocurrencies have taken a further step in the direction of commodities with the emergence of cold storage. Cold storage means generating and storing the crypto coins private keys in an offline environment in order to avoid the online vulnerability to hacking. The most popular cold storage options are: a) a paper wallet that contains a pair of private/public keys printed on a piece of paper, with keys generated offline securely; b) A hardware wallet, which is an electronic device, but the most robust cold storage choice for cryptos. The three popular options offered on the market are Ledger Nano, Trezor and Keepkey; c) a USB drive can also be a cold wallet, easy to obtain but with the risk that anyone having access to the USB can access the crypto.

At the same time, a platform called Coinbase where users can buy and sell Bitcoins and other cryptocurrencies is launching a program for institutional investors that will pay them a set interest rate for certain currencies likely to be between 5 and 8% - when they keep their coins in its custody service. Further evidence of the storability of physical crypto-assets.

We propose to extend to cryptocurrencies the convenience yield proposed by famous economists like Kaldor (1939), Working (1949) or Brennan (1958) to represent the virtue of owning a physical commodity as opposed to a derivative/financial contract written on it. We argue that holding physical Bitcoins presents a benefit because of illiquidity and frictions in the purchase of spot Bitcoins, hence that a convenience yield deserves to be introduced; note that established commodity traders use in the design of their strategies the convenience yield introduced in an academic setting (see Gabillon, 1991). We denote  $y$  this convenience yield (defined as benefit minus storage cost) and write the spot- forward relationship for cryptocurrencies, namely

$$f(t, T) = e^{(r-y)(T-t)}$$

Where  $f(t, T)$  denotes the price at date  $t$  of the Future contract maturing at date  $T$ ,  $S(t)$  the spot price of the crypto,  $r$  is the interest rate and  $y$  the convenience yield, both assumed constant over the period  $(0, T)$ . It is important to notice that this relationship as well prevails for currencies and equities, namely all asset classes that have been alternatively proposed to represent Bitcoins; it has the beautiful property of being independent of the choice of any given model for the spot price (see Geman, 2005)

The storability and spot-forward provide us with one easy way to short bitcoins and repeat below the replication argument (see Taleb, 2014) introduced in the famous papers of Black-Scholes (1973) and Merton (1973).

### III. PLAIN VANILLA OPTIONS ON BITCOINS

#### A. The General Setting

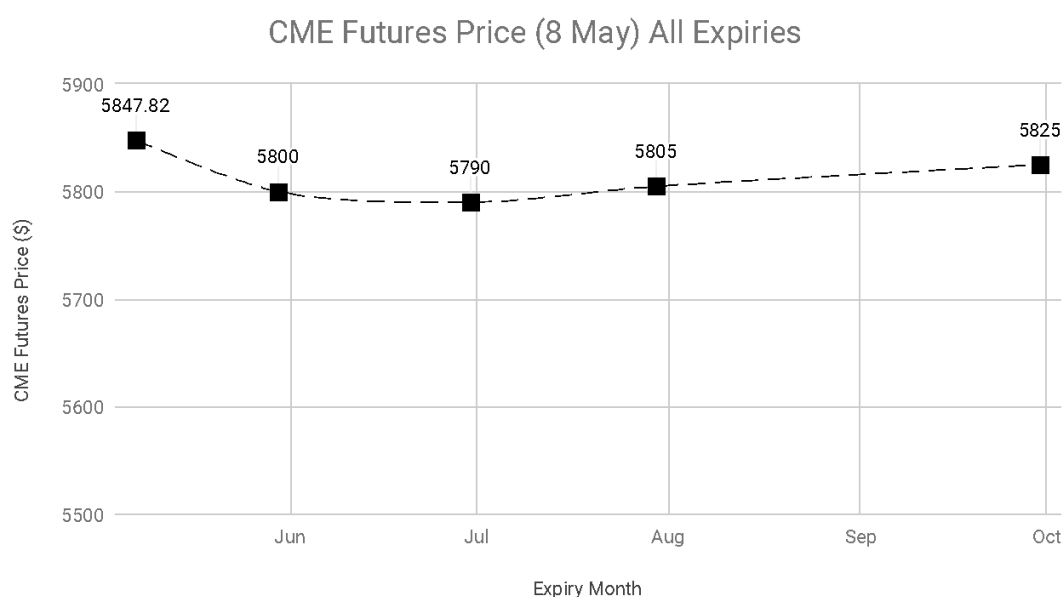
Options on Bitcoin offer buyers the usual benefits of options, calls giving, in particular, the positive exposure to the spot price without paying the full price of the underlying.

We argue in favour of a diffusion to represent the evolution of the cryptocurrencies, because of their unique property of being continuously traded, - erasing, in particular, the problems created by the overnight closure of equity markets. Moreover, we choose the geometric Brownian motion often proposed in the literature and an acceptable model for the average lifetime of traded options.

Hence, the price process ( $S(t)$ ) is driven under the risk-adjusted measure  $Q$  by the stochastic differential equation:

$$dS(t) = \alpha S(t)dt + \sigma S(t)d\hat{W}$$

where  $\hat{W}$  is a  $Q$  Brownian motion and the volatility is assumed constant in a first instance, then stochastic to account for the features displayed by the price trajectories and volatility pattern.



**Exhibit 4:** CME forward Curve on May 8, 2019

We provide in Exhibit 3 the forward curve observed on the CME on May 8, 2019. It was very mildly decreasing, showing that market participants did not have clear anticipations of a rise or decline of the cryptocurrency – this was before the project of the Facebook crypto became a real possibility. The forward curve provides us with the quantity  $(r - y)$ , which is our targeted risk-neutral drift and does not need to be broken into the two components, thus avoiding any numerical assumption on the proper risk-free rate for bitcoins.

Lastly, under constant interest rates - an assumption already made above and certainly acceptable for short-dated options, the forward price  $f(t, T)$  and the Future price (quantity observed on the Exchange) of the same maturity  $F(t, T)$  are equal.

The other key parameter is the volatility parameter in the diffusion process driving  $S(t)$ ; we propose to use the (annualized) volatility computed over  $m$  days of observation and given by the formula

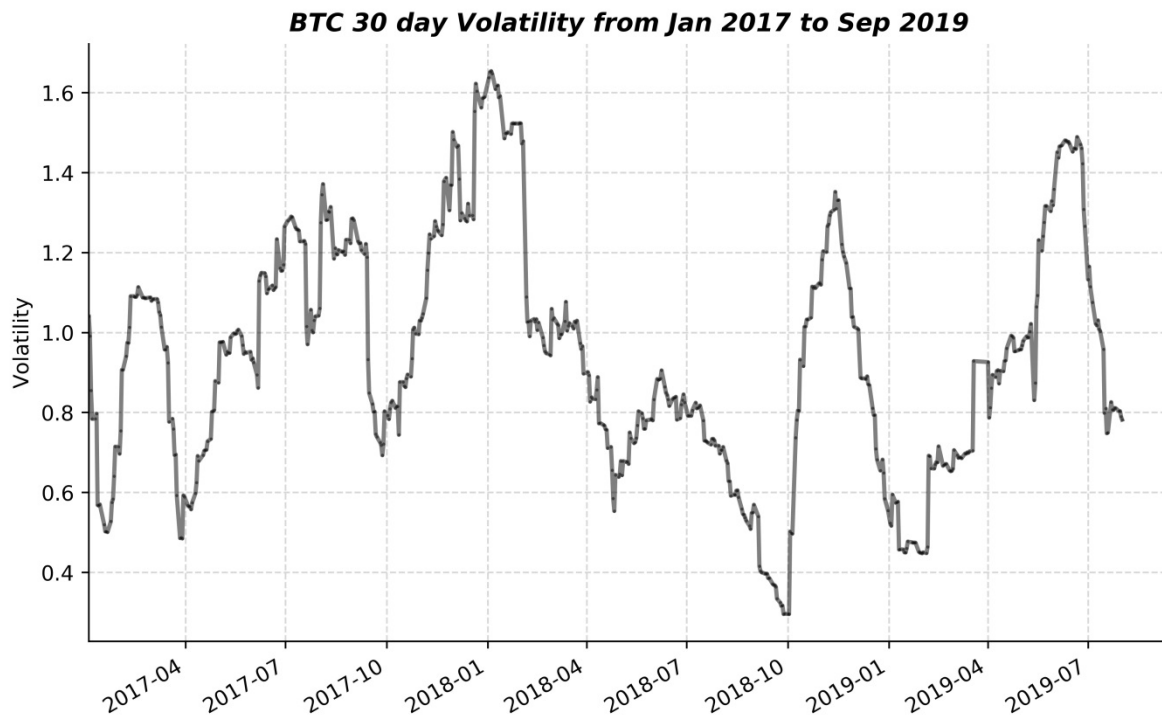
$$\sigma_{cc} = \sqrt{365} \cdot \sqrt{\frac{1}{N} \sum_{i=1}^N (r_i - \underline{r})^2}$$

since Bitcoin trades 365 days a year.

### B. Numerical Valuation of Cryptocurrency Options

Exhibit 5 shows that the volatility is not constant over the period May 2018 to May 2019 and has a pattern that is mean reverting to the level of 60%, with extreme values of 22% and 100%. Note that the highest values of the volatility were never observed in equities or ordinary commodities, but only in electricity markets in the early days of deregulation (see Geman and Vasicek, 2001) – interestingly, electricity is a continuously consumed commodity. This is an agreement with our view of Bitcoin as some extension of electricity electrons, and their status as a hybrid of equities and commodities as we argued before.

We consider as date 0 May 8, 2019. The spot price of Bitcoin was  $S(0) = \$5750$ . We choose for maturity date September 27, 2019, meaning that  $T = 142/365$  since Bitcoin traded every calendar day. The Future price for maturity T is \$5815 (see Exhibit 3), and we can derive from the spot-forward relationship the value  $x = 2.8\%$



**Exhibit 5:** 30-day volatility for Bitcoin.

We consider a strike K of \$6,000 and use the Merton (1973) formula established for stocks paying a continuous dividend and extended to a convenience yield (see Geman 2005).

$$C(S_t, t) = N(d_1) S_t e^{-y(T-t)} - N(d_2) K e^{-r(T-t)}$$

$$d_1 = \frac{1}{\sigma \sqrt{T-t}} \left( \ln \ln \left\{ \frac{S_t e^{-y(T-t)}}{K e^{-r(T-t)}} \right\} + \frac{1}{2} \sigma^2 (T-t) \right)$$

$$d_2 = d_1 - \sigma \sqrt{T-t}$$

## Plain Vanilla Option Prices

We assume in a first stage a volatility of 60% as it is an average value over the last year and the parameters mentioned above and display the option price below; the date of analysis is May8, 2019

**Example 1:**  $S = \$5750$ ,  $K = \$6000$ ,  $T - t = 0.386$  (141/365 days),  $r - y = 0.105$ ,  $\sigma = 0.6$

Vanilla Call Price: \$872

$S = \$5750$ ,  $K = \$6000$ ,  $T - t = 0.139$  (51/365 days),  $r - y = 0.105$ ,  $\sigma = 0.6$

Vanilla Call Price: \$448

Looking at the prices quoted on the website Deribit.com (accessed 8th May), we found a price of \$790 - \$825 (Bid - Offer) for the same initial date and expiry 27 Sep 2019 and \$429 - \$461 (Bid - Offer) for the same initial date and expiry 28 Jun 2019.

### *C. Volatility Uncertainty, Bitcoin Market Incompleteness and Option Price Bounds*

The property of increasing monotonicity of European calls and puts with respect to the volatility parameter as expressed by a strictly positive vega in the Black - Scholes model leads to the natural intuition that bounds on volatility provide market participants with bounds on option prices. This result is mathematically not straightforward – and does not hold if the bounds are themselves stochastic - is quite useful as volatility is not constant over the lifetime of the option in the case of Bitcoin (see Exhibit 4), as in all asset classes.

In their (1995) paper, Avellaneda et al formally addressed this issue when pricing and hedging options in an environment of uncertain volatility. Assuming that this volatility lies between bounds denoted by  $l$  and  $L$ , the authors validate our intuition that the option price is comprised between the numbers obtained for  $l$  and  $L$ . They also observe that  $l$  and  $L$  could be inferred from observed implied volatilities of option prices or from high-low peaks in historical volatilities. Accordingly, a rational agent would try to sell the option using the upper bound value of the volatility and buy at the lower bound value.

Returning to the example of the call analysed above, we may decide that two appropriate bounds for the volatility are 100% (\$794) for the upper bound and 40% (\$581) for the lower bound. There is a large difference both in absolute and relative numbers but emphasizes the current *incompleteness* of the bitcoin market. Still, the Black- Scholes formula holds for the bounds, because of the beautiful *robustness* of the Black-Scholes-Merton model. Note that we dispute the necessity of introducing jumps – which would add to incompleteness, as it happened in the credit market in 2008 - in the price trajectories proposed by some authors for bitcoin price trajectories; in this unique setting of continuous time trading, even complex tools of mathematical statistics would not discriminate between jumps added to the diffusion and stochastic volatility.

**Example 2:**  $S = \$5750$ ,  $K = \$6000$ ,  $T - t = 0.386$ ,  $r - y = 0.105$ ;  $\sigma = 0.4$  Vanilla Call Price: \$581

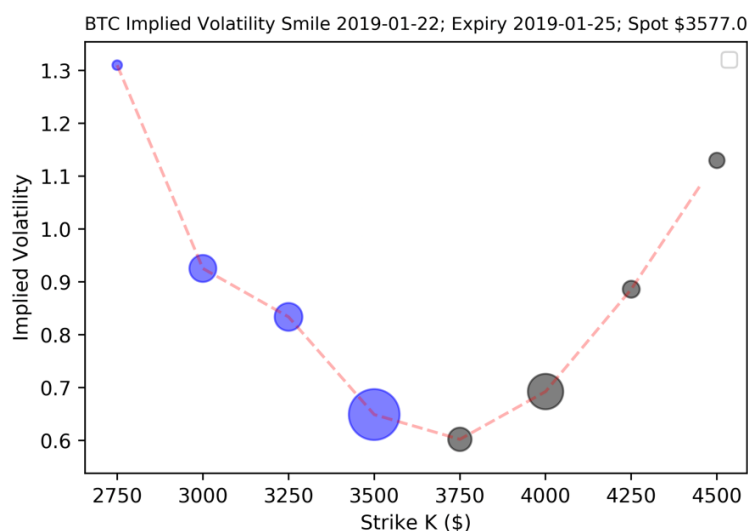
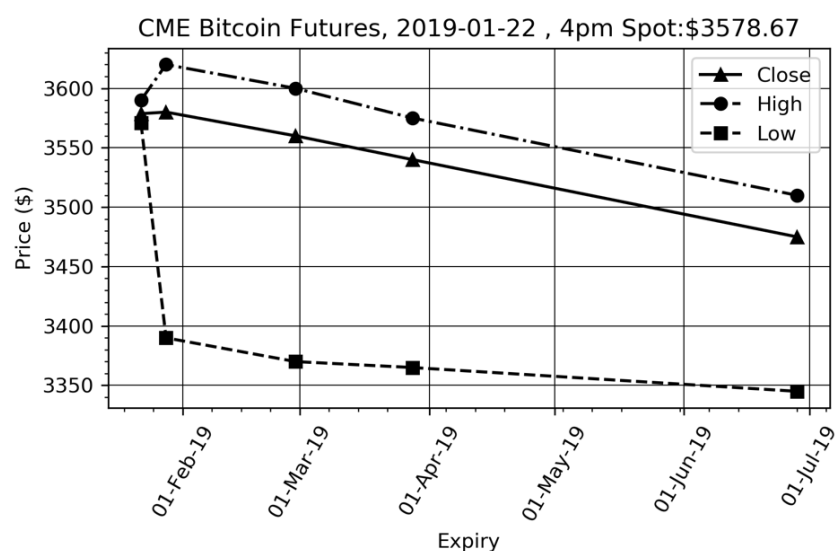
$S = \$5750$ ,  $K = \$6000$ ,  $T - t = 0.386$ ,  $r - y = 0.105$ ;  $\sigma = 1.0$

Vanilla Call Price: \$794

#### IV. SOME PROPERTIES OF BITCOIN FORWARD CURVES AND OPTIONS SMILES/ SKEWS

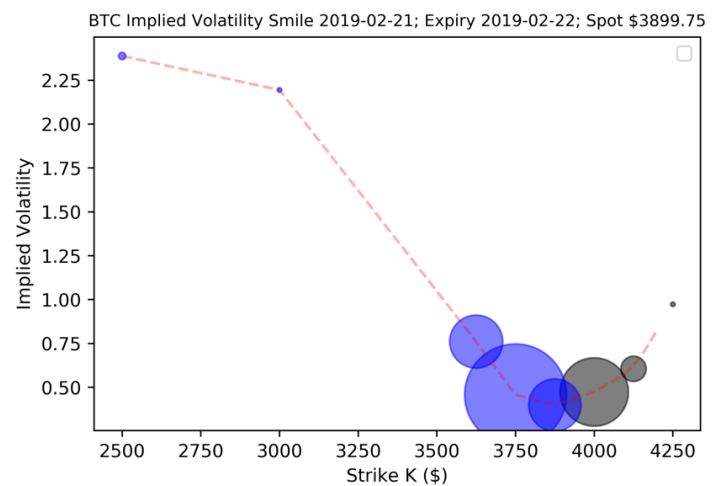
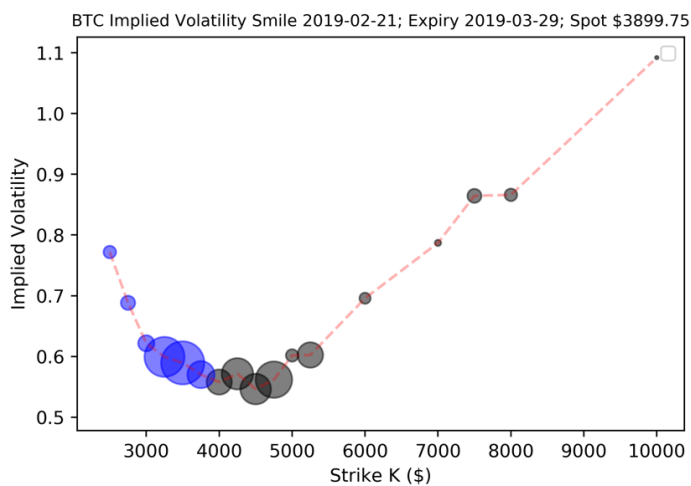
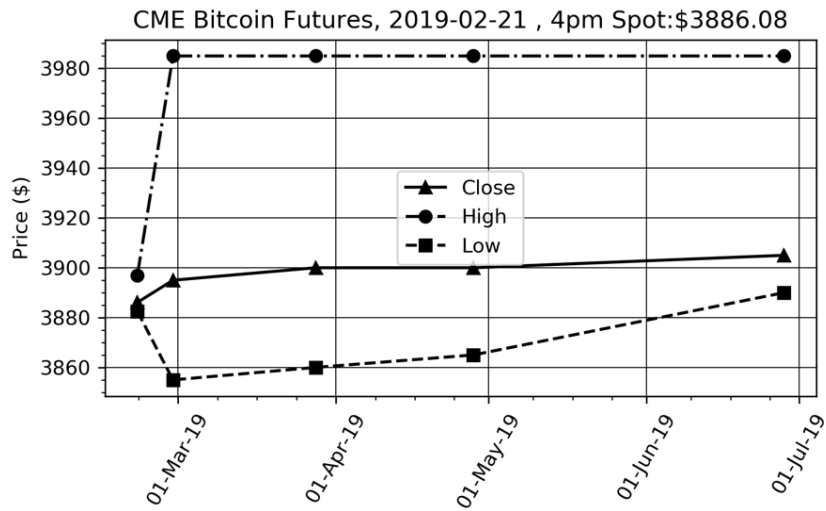
We use the classical rule of puts for strikes below the At the Money (ATM) value and calls for strikes above in order to plot the implied volatility extracted from option prices traded on the Deribit exchange as a function of the strike; the circles are proportional to traded volumes throughout all graphs and closing prices are used for the options.

a) On January 22, for maturity January 25, the smile is symmetric around the ATM value of \$3577. The three forward curves (high, close, low) are mildly declining, indicating that the market is not anticipating big moves of Bitcoin prices; the volatility is high however, comprised between 60% and 110%.



**EXHIBIT 5:** CME Futures curves and Deribit Implied Volatility Smile on 2019-01-22.

**b)** On February 21, for maturity February 22, essentially the same property prevails for the shape and the volume around the ATM price of \$3899, probably because of the 8.7% price increase since January 22. The three forward curves are increasing and 5 weeks to expiry the smile is skewed to the right, indicating that market participants are buying calls for price exposure.



**EXHIBIT 6:** CME Futures curves and Deribit Implied

Volatility Smile on 2019-02-21 for long and short expiry.

c) On April 4, the spot was at \$4939; the high forward curve is markedly increasing while the close and low are moderately decreasing. The neutral view of the investors is reflected in a fairly symmetric smile for the maturity of April 26, except for the large volume of puts with strike \$3500, which seems to indicate that a large long position was buying out-of-the-money puts on that day.

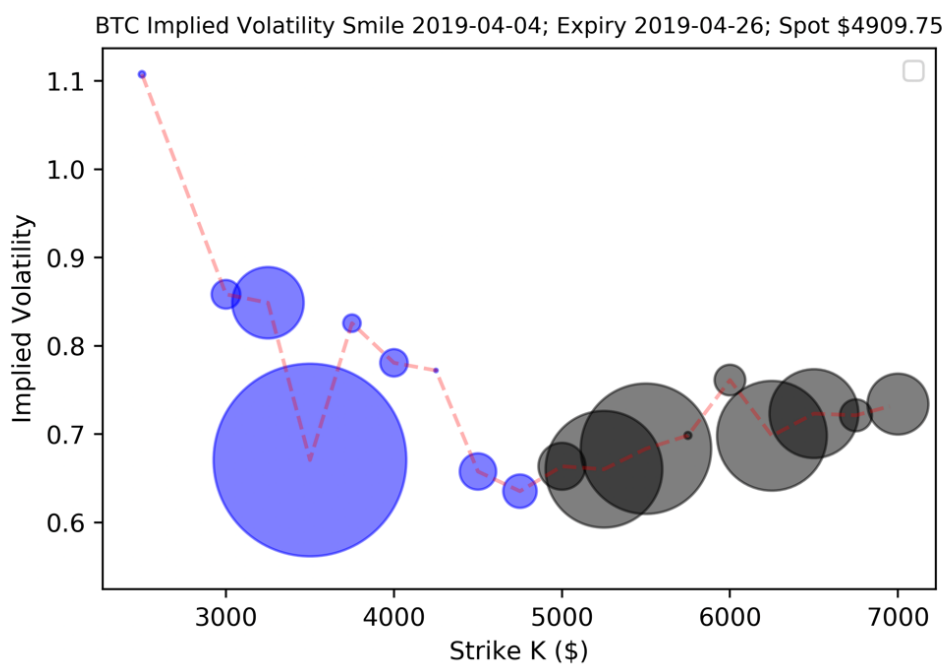
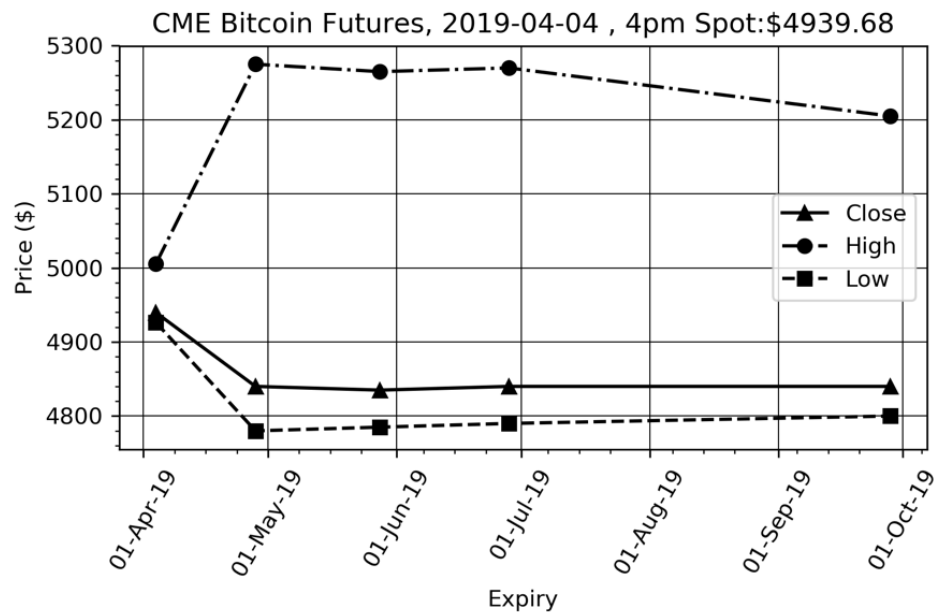
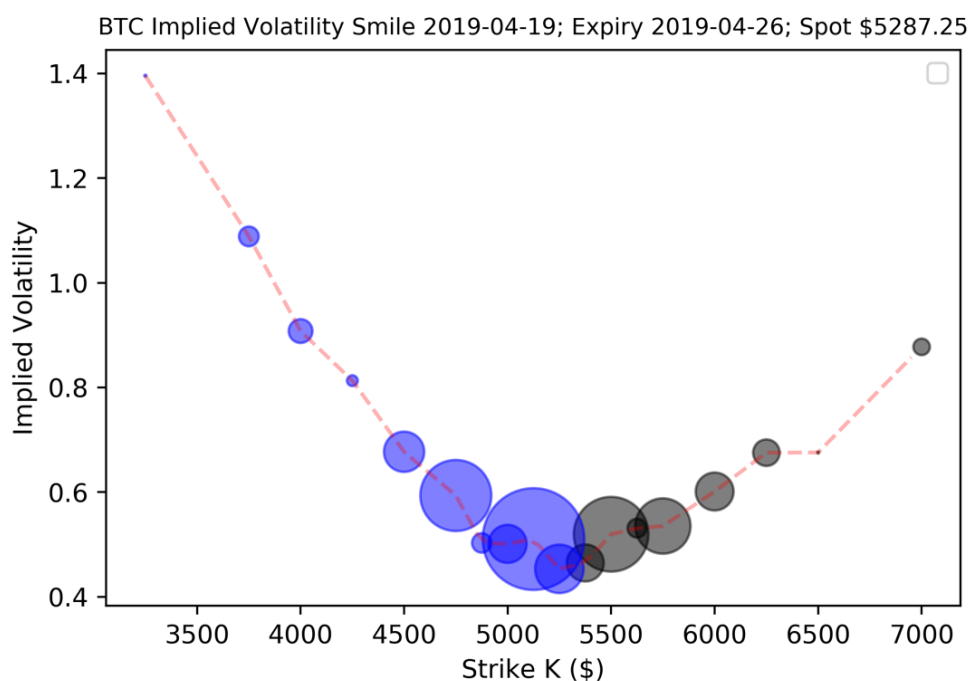
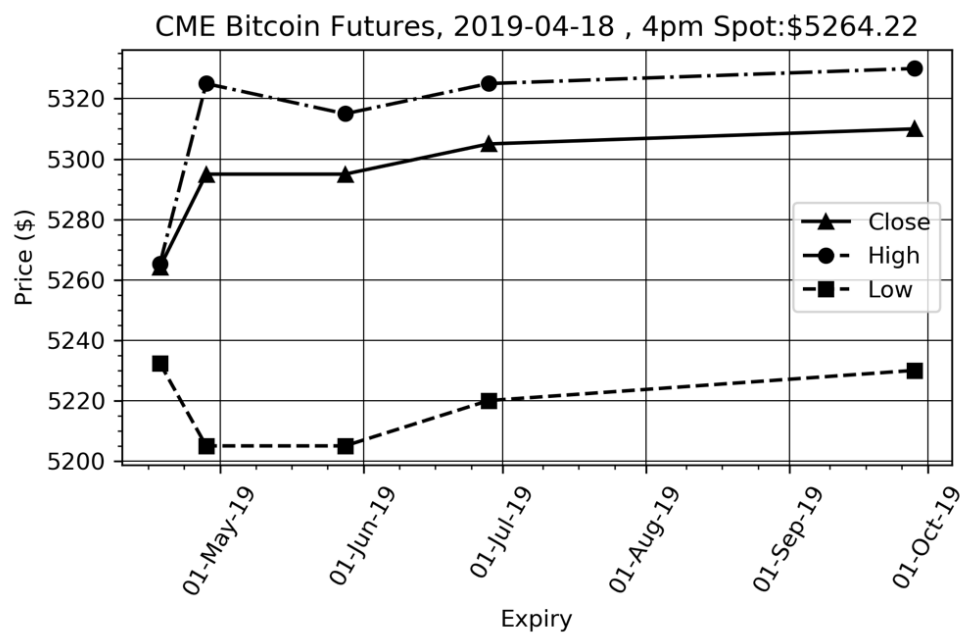


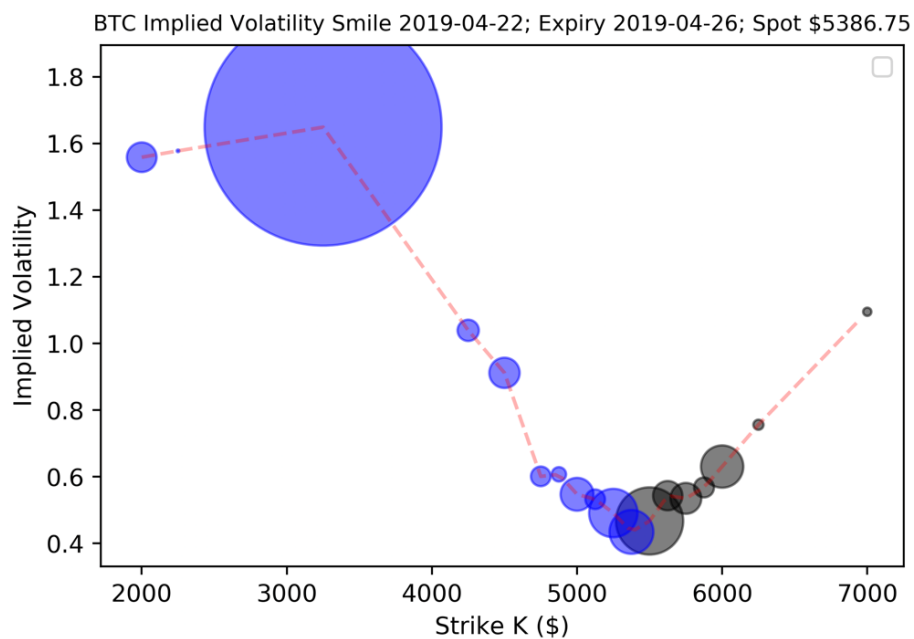
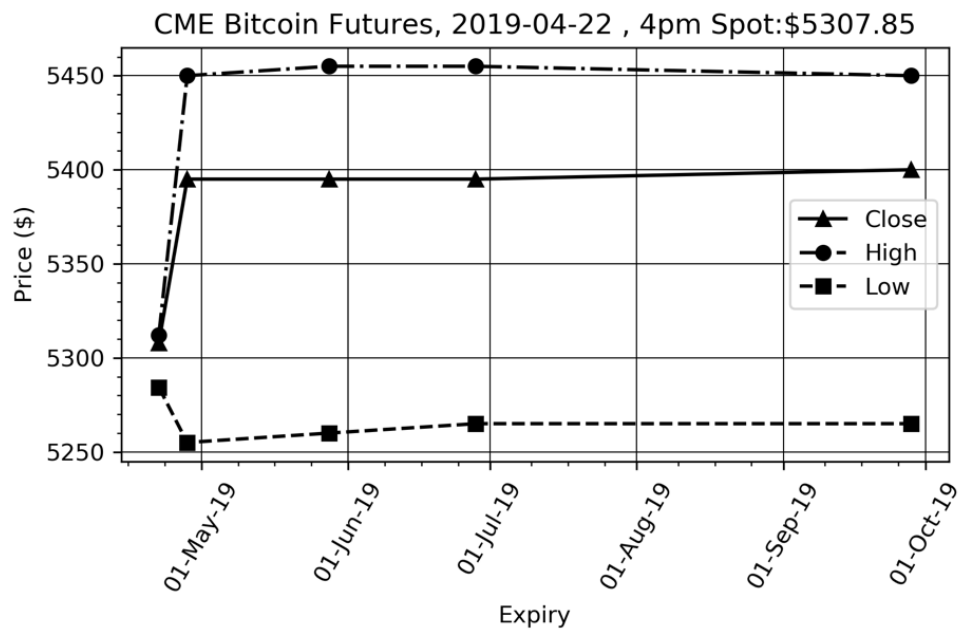
EXHIBIT 7: CME Futures curves and Deribit Implied Volatility Smile on 2019-04-04.

d) On April 19 for a time to maturity of 7 days we observe again a remarkably symmetric smile and prudently conclude that close to maturity, the smile is fairly symmetric, indicating that investors do not generally anticipate a big collapse of Bitcoin prices within one or two days. The three curves were increasing. Note that the period 19-22 of April, the CME was closed for Good Friday; the Deribit Exchange however, is open every day.



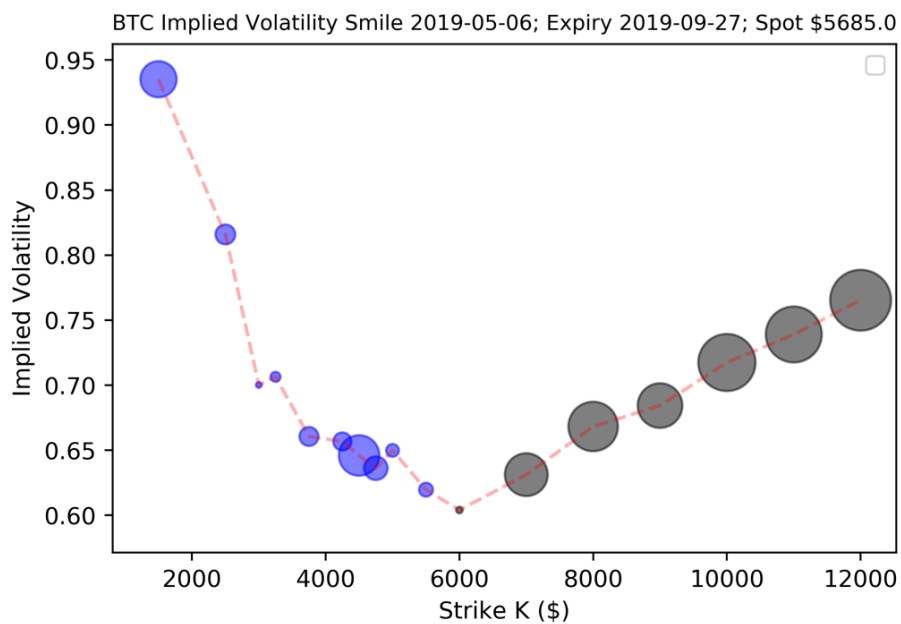
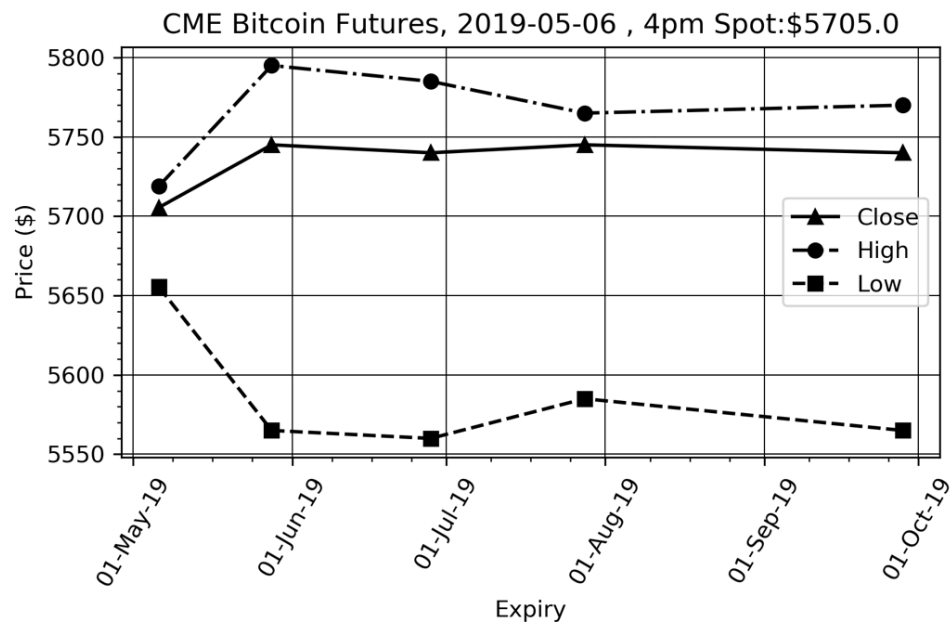
**EXHIBIT 8:** CME Futures 2019-04-28 curves and Deribit Implied Volatility Smile on 2019-04-19

e) On April 22, the spot has gone up to \$5307; the investors are clearly optimistic as two forward curves out of three are increasing. The volatility smile for a two-day duration is fairly symmetric as observed earlier for very short dated options; a large put buyer disrupts the symmetry with an implied volatility of 160% compared to a 40% volatility at-the-money



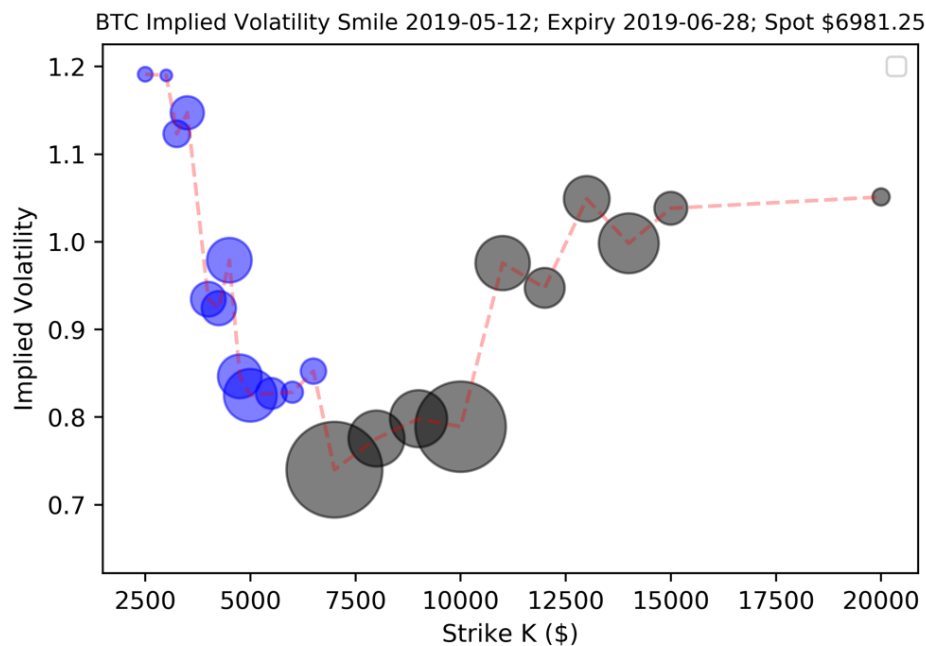
**EXHIBIT 9:** CME Futures curves and Deribit Implied Volatility Smile on 2019-04-22

f) On May 6, the spot is higher at \$5705; the investors continue to be positive as the high and close forward curves are increasing and the low one is flat at the spot level. Regarding the options maturing on September 27, the smile is clearly to the right, with a much larger volume of calls and investors betting on more price increases by buying calls at volatilities comprised between 62% and 75% for a strike of \$12,000.



**EXHIBIT 10:** CME Futures curves and Deribit Implied Volatility Smile on 2019-05-06.

g) Turning to the date of May 12, the spot price had gone up to \$6981 from \$5319 on May 5, meaning an increase of 28%. Remarkably, the liquidity was essentially located in calls - probably buyers who had not bought Bitcoins earlier -, and the implied volatilities were higher, ranging from 70% to 100% for the calls and 120% for the puts. Note that on May 6 the ATM volatility was lower at 60%.



**EXHIBIT 11:** Deribit Implied Volatility Smile on 2019-05-12.

We can conclude from this section that there is indeed consistency between the forward curves and option smiles/skews. It also appears that the shapes observed for the latter ones at this point in time bring Bitcoins closer to the asset class of equities than any other asset class.

## V. CONCLUSION

We have investigated in this paper a number of bitcoin properties, in particular the storability and convenience yield, which allows us to write the valuable spot-forward relationship. This one allows us to identify the risk-neutral drift in a Black-Scholes setting, whose robustness provides us bounds on option prices. We extract from options traded on the Deribit exchange smiles and skews that we compare to forward curves on the same dates of analysis. Even though large discussions are taking place around Libra and the proposal by Facebook of introducing in its community as a new means of payment, it seems that market participants still view Bitcoin as an investment vehicle and the options written on it as a way to get exposure or protection.

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